INAM ULLAH KHAN

AI/ML Engineer

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AI/ML Engineer building production systems with LLMs, RAG, and deep learning. Experienced in scalable ML applications, medical AI platforms, RESTful APIs, and applied AI research. Skilled in Generative AI, computer vision, and deep learning with 7 Q1 journal publications.

EXPERIENCE

Ethics Advance Technology Ltd. Aug 2025 – Present
AI/ML Engineer *Dhaka, Bangladesh*

- Developed RESTful APIs using **FastAPI** to support LMS functionalities and integrate AI-powered features.
- Integrated domain-adapted **LLMs** with prompt engineering, document chunking, embeddings, vector databases, and **RAG** pipelines for intelligent responses.
- Deployed LLM-powered LMS solutions into production with real-time API endpoints and web application integration.

United International University Jun 2023 – Jul 2024
Research Assistant *Dhaka, Bangladesh*

- Designed lightweight **Compact Convolutional Transformer (CCT)** architectures, reducing computational complexity while maintaining high accuracy.
- Developed factual datasets and performed **model ranking and evaluation** to assess accuracy and consistency.
- Conducted research in **medical AI, computer vision, and deep learning**, publishing 7 papers in Q1 journals including IEEE Access, MDPI Biomedicines, and Neural Computing and Applications.

PROJECTS

- **Bangladesh Medical Education LMS (LLM-Based)**
  FastAPI · LangChain · OpenAI API · RAG · Vector DB
  Developed a medical Learning Management System using LLMs and RAG for accurate retrieval from clinical documents, supporting multi-turn conversations with context awareness.
- **AI Healthcare Triage System**
  LLMs · Prompt Engineering · FastAPI · REST APIs · AI Agents
  Built an LLM-powered healthcare triage system that processes patient information and symptoms through conversational AI workflows, generates structured clinical assessments, and supports intelligent case routing via FastAPI-based REST APIs.
- **Radiology Annotation Dashboard**
  Python · Dash · Plotly · GPT-4V · DICOM
  Built a production-ready medical image annotation platform supporting DICOM/PNG/JPG formats with AI-assisted labeling and interactive visualization.
- **X-Ray Analyzer** GitHub
  Python · Deep Learning · Computer Vision · X-Ray Analysis
  Developed an AI-based X-ray analysis system for automated medical image processing and disease prediction using deep learning and computer vision techniques.
- **Eye Disease Prediction System**
  FastAPI · TensorFlow · React
  Developed a full-stack deep learning application for retinal disease classification with optimized inference performance.

More Projects: Visit My GitHub

SKILLS

- **Programming:** Python, C, C++, JavaScript, SQL
- **AI/ML:** Machine Learning, Deep Learning, LLMs, Transformers, NLP, PyTorch, TensorFlow, Hugging Face
- **Generative AI:** RAG, AI Agents, Prompt Engineering, OpenAI API, LLM Integration, Embeddings, Semantic Search
- **AI Frameworks:** LangChain, FastAPI, Streamlit, React, Next.js, SQLAlchemy
- **Computer Vision:** YOLO, OCR, DICOM, Grad-CAM, Medical Imaging, Image Classification
- **Vector & Retrieval:** Qdrant, FAISS, PgVector, Vector Databases, Vector Search, Document Chunking
- **Databases:** PostgreSQL, MySQL, MongoDB, Supabase
- **Backend & APIs:** FastAPI, JWT, API Integration, Uvicorn
- **Tools:** Git, GitHub, GitLab, Pandas, OpenCV

EDUCATION

- **Daffodil International University** Dhaka
  B.Sc. in CSE 2019 – 2023
  CGPA: 3.63/4.00

CERTIFICATIONS

- **Youth Social Leadership Training** — International Center for Global Skill, Funded by USAID & New York University (2020)
- **Problem Solving (Basic)** — HackerRank (2023)
- **Oral Presentation Certificate** — 10th International Conference on Computer and Communications Management (ICCCM 2022), Japan
- **ICPC Participation** — International Collegiate Programming Contest (2021, 2022)

PUBLICATIONS

- **Modified CCT for Lung Disease Classification** — Intelligent Systems (2022) — *Modified CCT for efficient lung disease classification. [Q1]*
- **CAD System for Diabetic Retinopathy Using Modified CCT** — MDPI Biomedicines (2023) — *Computer-aided diagnostic system using modified CCT. [Q1, IF: 4.0]*
- **Lightweight Deep Learning for Diabetic Retinopathy** — IEEE Access (2023) — *Lightweight deep learning for retinal disease detection with reduced computational cost. [Q1, IF: 3.9]*

More Publications: Google Scholar Profile